Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Net force

Meaning in my words How would you explain net force to a friend?	**How can you tell?** Two people push one cart in opposite directions. Why do the arrow directions matter?
Example or sketch One person pushes 6 N right and another 4 N left. What single arrow replaces both?	**Non-example and why** A student adds those pushes to get 10 N right. Which push did they count the wrong way?

Newton’s second law

Meaning in my words What does Newton’s second law tell us about a cart?	**How can you tell?** Why add the forces before dividing by mass?
Example or sketch A 2 kg cart feels 7 N right and 3 N left. What acceleration do you predict?	**Non-example and why** A partner uses 7 N alone to find acceleration. Which part of the diagram contradicts that?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Direction

Meaning in my words	How can you tell?
How would you explain direction to a friend?	A cart moves right while its net force points left. Which way does acceleration point?
Example or sketch	**Non-example and why**
Draw arrows for a cart moving left and slowing down.	"The biggest force points east, so the cart must be moving east." What can we actually tell if the net force is east?

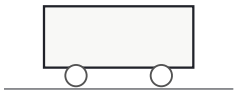
Balanced

Meaning in my words	How can you tell?
What does it mean to say the forces on a cart are balanced?	A cart feels 5 N left and 5 N right. How can you tell its horizontal forces balance?
Example or sketch	**Non-example and why**
A book rests on a table. What two vertical arrows would you draw?	One cart pushes another, and the second pushes back. Do these two forces balance on the first cart?

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Draw an object with two opposing forces. Write the signed sum under the net arrow.



My marks mean

Moving left. Pushed right.

13

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Distinguish motion, net force, and acceleration.

CHOOSE ONE JOB TOGETHER

CONTROLS ENGINEER ☐

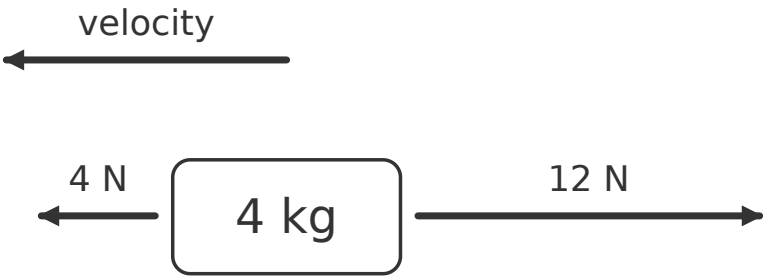
Draw the net-force and acceleration arrows. Tell the operator what happens to speed while the body still moves left.

REPLAY ANALYST ☐

Label separate arrows for motion and acceleration. Explain to viewers how they can point opposite ways.

Point to a clue. Keep your first marks; add the update using the color key.

A tracking model moves left. Its **4 kg** body feels **12 N** right and **4 N** left. The status display has separate motion and force arrows.



01 OUR FIRST CALL

What happens to speed at first? Include the net force and acceleration in your call.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Check the updated display. What does a force arrow tell you, and what does it not tell you?

Switch speakers. Can motion and acceleration point in opposite directions?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks